



Beneficial effect of boron in layered sodium-ion cathode materials – The example of $\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$

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HIGHLIGHTS

- Boron doping promotes the generation of multi-phase $\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$ layered material.
- $\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$ shows highly improved reversible cycling between 4.0 and 2.0 V vs. Na/Na^+ .
- Boron doping of layered manganese oxides results in high specific capacities.

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ABSTRACT

Sodium-ion batteries are regarded as a complementary drop-in technology to lithium-ion batteries because they promise lower cost and a higher degree of environmental friendliness. Among other reasons, these benefits come from the use of manganese-based materials, whose stabilization via cation substitution is intensively studied to improve the electrochemical performance. Although multiple elements have been considered as substituent, surprisingly, boron has not been reported for layered sodium-ion cathode materials up to date. Our investigation of layered $\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$ reveals an unexpectedly good electrochemical performance, with charge and discharge capacities of more than 175 mAh g^{-1} at 10 mA g^{-1} and 135 mAh g^{-1} at 500 mA g^{-1} . The measured capacities are among the highest ever reported for sodium-based layered oxides in the potential range of 4.0–2.0 V vs. Na/Na^+ .

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1. Introduction

To support the transformation of the energy infrastructure to a more sustainable and decentralized system, batteries are considered to be a crucial element. The storage of electricity in electrochemical devices is independent of the geographical location, scalable, increasingly cost-efficient and, therefore, fulfills the most important criteria for our current and future needs. However, the growing demand for electrochemical storage is raising concerns about the sufficiency of resources for currently used battery technologies and ecological risks associated with the raw materials used therein.

Although investigated since the 1970s, sodium-ion batteries (SIBs) have been more intensively studied in the past few years. They share the same working principle with lithium-ion batteries [1,2], but with some advantages attributed to the more abundant and cheaper sodium raw materials and the possible use of aluminum as negative current collector instead of the heavier and more expensive copper [3]. To strengthen these cost advantages, efforts are made to avoid the use of toxic elements, e.g., cobalt and nickel, in order to finally make sodium-ion batteries cheap and environmentally friendly. However, due to their expected lower energy performance with respect to LIBs, SIBs are especially considered for applications in large-scale energy storage devices as, e.g., power-grid buffering.

A large fraction of the research on SIBs is focused on cathode materials, for which two major categories can be identified: metal oxides and polyanionic compounds.

The use of sodium-intercalated layered metal oxides for electrochemical energy storage was reported in 1981 for the first time,

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mainly dealing with structural investigations of Na_xCoO_2 [4]. Other studies in the 1980s proved the feasibility as positive active materials of several Na_xMnO_2 compounds, whose syntheses and structures had been reported before in the early 1970s [5,6]. Since then, numerous studies investigated, and optimized, their electrochemical characteristics, often by the partial substitution of manganese with other elements, both, electrochemically active and inactive. These latter decrease the amount of active species in the compound, but usually enhance the stability of the material, as in the case of titanium, copper and magnesium [2]. Active substituents contribute to the electrochemical redox process themselves, as in the case of iron, nickel and cobalt [2]. However, in addition to the specific capacity values, the effects of the partial Mn substitution are numerous. Works often focused on the suppression of the Jahn-Teller distortion of manganese [7], in addition to the increase of capacity and average working potential [8] and cycling stability [9]. So far, quadruple, triple, dual, single and not-substituted manganese-based layered oxides have been reported [10] (see Table S1 in the supplementary information (SI)).

Optimally, the substituent(s) should be light in order to enable the increase (if active) or just a minor decrease (if inactive) of specific capacity. For that, aluminum and boron are perfect candidates since both elements are lightweight and, additionally, cheap, abundant and of low toxicity.

Aluminum doping has been already investigated in layered oxides (e.g., $\text{Na}_x\text{Al}_{0.1}\text{Mn}_{0.9}\text{O}_2$, $\text{Na}_{0.67}\text{Mn}_{0.65}\text{Ni}_{0.15}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ and $\text{Na}_{0.6}\text{Ni}_{0.22}\text{Al}_{0.11}\text{Mn}_{0.66}\text{O}_2$) [11–13] and polyanionic compounds (e.g., $\text{Na}_3\text{Al}_y\text{V}_{2-y}(\text{PO}_4)_3$) [14]. The use of boron in sodium-based layered oxides was also reported before, although only once and not for SIB applications. In fact, Velikokhatnyi et al. synthesized a boron-containing layered sodium manganese oxide for applications in the field of LIBs, i.e., making use of Na-Li ion exchange [15]. The effect of boron in SIB cathode materials was only investigated for the polyanionic compound $\text{Na}_3\text{V}_2(\text{P}_{1-x}\text{B}_x\text{O}_4)_3$, which exhibited an increased structural stability and enhanced electrochemical performance [16]. The limited research on the effects of boron is rather surprising because this element is frequently used as dopant in industrial and academic research, including, e.g., doping of carbonaceous anode materials for electrochemical energy storage [17].

Herein we present the investigation of the use of boron as substituent in layered $\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$. By using a solid-state reaction, we kept the synthesis scalable and facile [18] and discovered irregular but positive effects on the electrochemical performance.

2. Experimental

The active materials were synthesized as it follows. Stoichiometric ratios of sodium carbonate monohydrate (Sigma-Aldrich, $\geq 99.5\%$), manganese (II) acetate tetrahydrate (Sigma-Aldrich, $\geq 99\%$), and boric acid (Alfa Aesar, $99+\%$) or aluminum 2-ethylhexanoate (Strem, $\geq 96\%$) were ball milled for 90 min at 800 rpm using a zirconia jar and balls (ball to powder ratio: 1.7, 1.8, 1.6 for $\text{Na}_{2/3}\text{MnO}_2$, $\text{Na}_{2/3}\text{Al}_{0.11}\text{Mn}_{0.89}\text{O}_2$ and $\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$, respectively). The resulting materials were annealed at 773 K for 4 h in air in a muffle furnace (heating rate 5 K min^{-1}). After cooling to room temperature in the furnace, the materials were ball milled using the same milling parameters as before but with a ball to powder ratio of about 1.8 for all three materials, pressed into pellets and annealed at 1173 K for 6 h (heating rate 5 K min^{-1}). After cooling to room temperature (temperature profile shown in SI (Figure S7)), the pellets were ground with mortar and pestle and, finally, screened over a $45\text{ }\mu\text{m}$ sieve. In order to compensate the loss of sodium during the synthesis, excess amounts of this element

(0.75 eq. per mole $\text{Mn}_{0.89}\text{X}_{0.11}\text{O}_2$ with $\text{X} = \text{Mn, Al and B}$) were used to obtain a final $\text{Na}/(\text{Mn}:\text{X})$ ratio of about $2/3$ [3,19]. The three materials were characterized and tested without any further treatment. X-ray diffractograms (XRD) were recorded in Bragg-Brentano geometry in the 2θ range of $10\text{--}90^\circ$, using a step size of $0.01^\circ\ 2\theta$ and a time per step of 3 s on a Bruker D8 Advance diffractometer equipped with a CuK_α radiation ($\lambda = 0.154\text{ nm}$). The morphology of the active materials was investigated using scanning electron microscopy (SEM, Zeiss Auriga[®], samples sputtered with 4 nm of Pt) with an acceleration voltage of 10 kV.

Electrodes were made mixing each of the active materials with 10 wt% solution of polyvinylidene difluoride (PVdF, Solef 6020, Arkema) in *N*-Methyl-2-pyrrolidone (NMP, Sigma-Aldrich, anhydrous, 99.5%) and conductive carbon black (Super C65, IMERYS) via ball milling. The slurries (solid content 29 wt%) were casted on aluminum foil ($20\text{ }\mu\text{m}$ thick) with a doctor blade (wet coating thickness of $80\text{ }\mu\text{m}$). The coated foils were pre-dried at 333 K in air for 24 h. Disk electrodes ($\varnothing = 12\text{ mm}$) were cut, dried at 393 K under vacuum for 24 h, pressed (7 t cm^{-2}), dried again at 393 K under vacuum for 24 h, and, finally, stored in an Ar-filled glovebox. The dry electrode composition was 85 wt% active material, 10 wt% carbon black and 5 wt% binder. The average active material mass loading was around 2.5 mg cm^{-2} .

Three-electrode cells were assembled using sodium metal (99.8% , Acros Organics) as the counter and reference electrodes. Whatman[®] GF/D glass fiber was used as separators, drenched with 300 μL of filtered electrolyte, i.e., the solution of NaPF_6 (sodium hexafluorophosphate, Alfa Aesar, $99+\%$) in PC (propylene carbonate, BASF, 99.95%). Prior to use, the electrolyte was filtered with a $0.45\text{ }\mu\text{m}$ pore size polytetrafluoroethylene syringe filter. The resulting electrolyte solution stayed clear even after several weeks of storage in the glove box. The NaPF_6 concentration in the electrolyte was measured via inductively coupled plasma optical emission spectrometry (ICP-OES), yielding sodium and phosphorus concentrations of 0.93 and 0.94 mol L^{-1} , respectively. No significant elemental impurities were detected. The conductivity and viscosity of the electrolyte were 6.37 mS cm^{-1} and 7.46 mPa s at 293 K, in accordance with previous studies [20]. Electrolyte preparation and cell assembly were carried out in a glove box (MBraun) with oxygen and water contents below 0.1 ppm.

Galvanostatic cycling tests were carried out on a Maccor battery tester 4300 and cyclic voltammetry on a VMP3 Potentiostat (Biologic). All electrochemical tests were carried out in climatic chambers at a temperature of $293 \pm 1\text{ K}$. All potential values given in this manuscript refer to the Na/Na^+ quasi-reference couple.

3. Results and discussion

ICP-OES results confirm the overall composition of the prepared materials to be near to the targeted overall values: $\text{Na}_{0.67}\text{Mn}_{1.00}$ (N_M), $\text{Na}_{0.66}\text{Al}_{0.11}\text{Mn}_{0.89}$ (NAM) and $\text{Na}_{0.64}\text{B}_{0.11}\text{Mn}_{0.89}$ (NBM).

According to the determined material compositions, two different theoretical capacity values can be calculated for each compound by Faraday's law (see Table 1). The first value refers to the complete removal of sodium (i.e., $2/3$ eq. of Na^+). Since Al and B are electrochemically inactive, the theoretical specific capacity for maximum sodium removal decreases (0.11 eq. Na^+ less can be extracted).

The second theoretical capacity value is based on the Mn redox reaction (from Mn^{3+} to Mn^{4+} and vice versa). In this case, the theoretical specific capacity is expected to decrease by about 8% for NAM and 7% for NBM with respect to N_M. Nonetheless, the complete and reversible sodiation of the materials is not achievable in practice because of material degradation. Thus, the substitution strategy may still result in higher practical capacities if the resulting

Table 1

Composition, molar mass and calculated theoretical specific capacities of the investigated materials.

Abbreviation	Composition	Molar Mass (g mol^{-1})	Theoretical Capacity upon Na^+ Extraction (mAh g^{-1})	Theoretical Capacity for $\text{Mn}^{3+}/\text{Mn}^{4+}$ (mAh g^{-1})
N_M	$\text{Na}_{2/3}\text{MnO}_2$	102.3	176	262
NAM	$\text{Na}_{2/3}\text{Al}_{0.11}\text{Mn}_{0.89}\text{O}_2$	99.3	151	240
NBM	$\text{Na}_{2/3}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$	97.5	154	244

material is stabilized upon sodiation/desodiation.

The SEM micrographs of the three compounds show a rather inhomogeneous primary particle size distribution (see Fig. 1). Also, N_M and NAM show less structured particles in contrast to NBM, which shows flake-like particles with a clearly visible layered structure at higher magnification (Fig. 1f).

XRD patterns of the materials are depicted in Fig. 2. The less intense signals are in the focus here, while the complete patterns and exemplary Rietveld refinements are reported in the SI. The diffractograms of all materials show the coexistence of several different phases as evidenced via the comparison with reference materials. Signals in the range of $12\text{--}14^\circ 2\theta$ are correlated with water intercalation from air [21]. Furthermore, it is important to note that no evidence of crystalline boron- or aluminum oxides is found. It should be noted that the presence of the different phases in the materials is additionally confirmed by Rietveld refinement.

N_M is mainly composed of two phases, the hexagonal (main component ($\sim 86\%$), space group $P6_3/mmc$; PDF: 00-027-0751) and the orthorhombic (side component ($\sim 14\%$), space group $Cmcm$; PDF: 00-027-0752). NAM is mainly composed of the same phases but with a different ratio (hexagonal phase $\sim 93\%$ and orthorhombic phase $\sim 7\%$).

In contrast, NBM consists of multiple phases which could not be identified to full extent. However, several phases might be present as detected from the visual comparison of the reflections with reported compounds, particularly two monoclinic phases (space group $C2/c$ [7] and space group $C2/m$; PDF: 01-075-324), one

hexagonal phase (side component, space group $P6_3/mmc$) and one orthorhombic phase (side component, space group $Cmcm$). Nevertheless, the structural effects of boron doping deserve a more detailed investigation by other techniques such as Fourier transform infrared spectroscopy (FTIR), electron paramagnetic resonance (EPR) and neutron diffraction.

Summarizing, all materials show different phase compositions, although they have been synthesized in identical temperature conditions. Indeed, it is known that sodium manganese oxides form various layered structures depending on the sodium content (e.g., 0.58, 0.70 or 1.00), annealing conditions and degree of Jahn-Teller distortion [6,22–25]. It is also important to remark that the coexistence of several phases has been reported very beneficial for the electrochemical performance of, e.g., $\text{Na}_{0.76}\text{Mn}_{0.5}\text{Ni}_{0.3}\text{Fe}_{0.1}\text{Mg}_{0.1}\text{O}_2$ [8].

The electrochemical performance of the three different materials was investigated by cyclic voltammetry and galvanostatic cycling within different potential ranges.

In particular, cyclic voltammetry (CV) measurements were performed at 0.1 mV s^{-1} within $4.6\text{--}1.5 \text{ V}$ or $4.0\text{--}2.0 \text{ V}$ vs. Na/Na^+ . The cyclic voltammograms depicted in Fig. 3 show multiple peaks for each compound, although only manganese is expected to be electrochemically active. The reasons for this commonly observed behavior are the sodium vacancy ordering in the sodium layers, phase transitions upon (de-)sodiation, and the coexistence of different phases, each comprising different sodium sites [3,8]. For N_M (see Fig. 3a), the CV in the range of $4.0\text{--}2.0 \text{ V}$ is similar to the

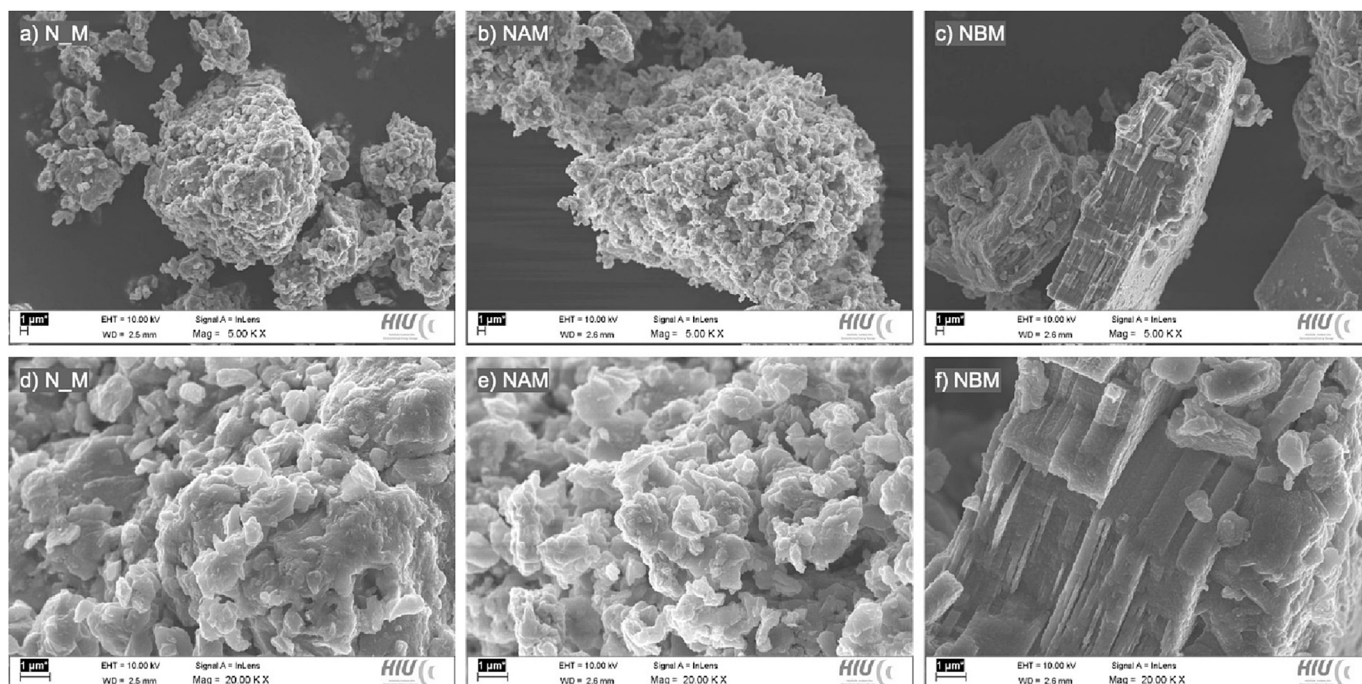


Fig. 1. SEM micrographs at 10 kV and 5 kx (a–c) or 20 kx (d–f) of $\text{Na}_{0.67}\text{MnO}_2$ (N_M, a and d), $\text{Na}_{0.66}\text{Al}_{0.11}\text{Mn}_{0.89}\text{O}_2$ (NAM, b and e) and $\text{Na}_{0.64}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$ (NBM, c and f).

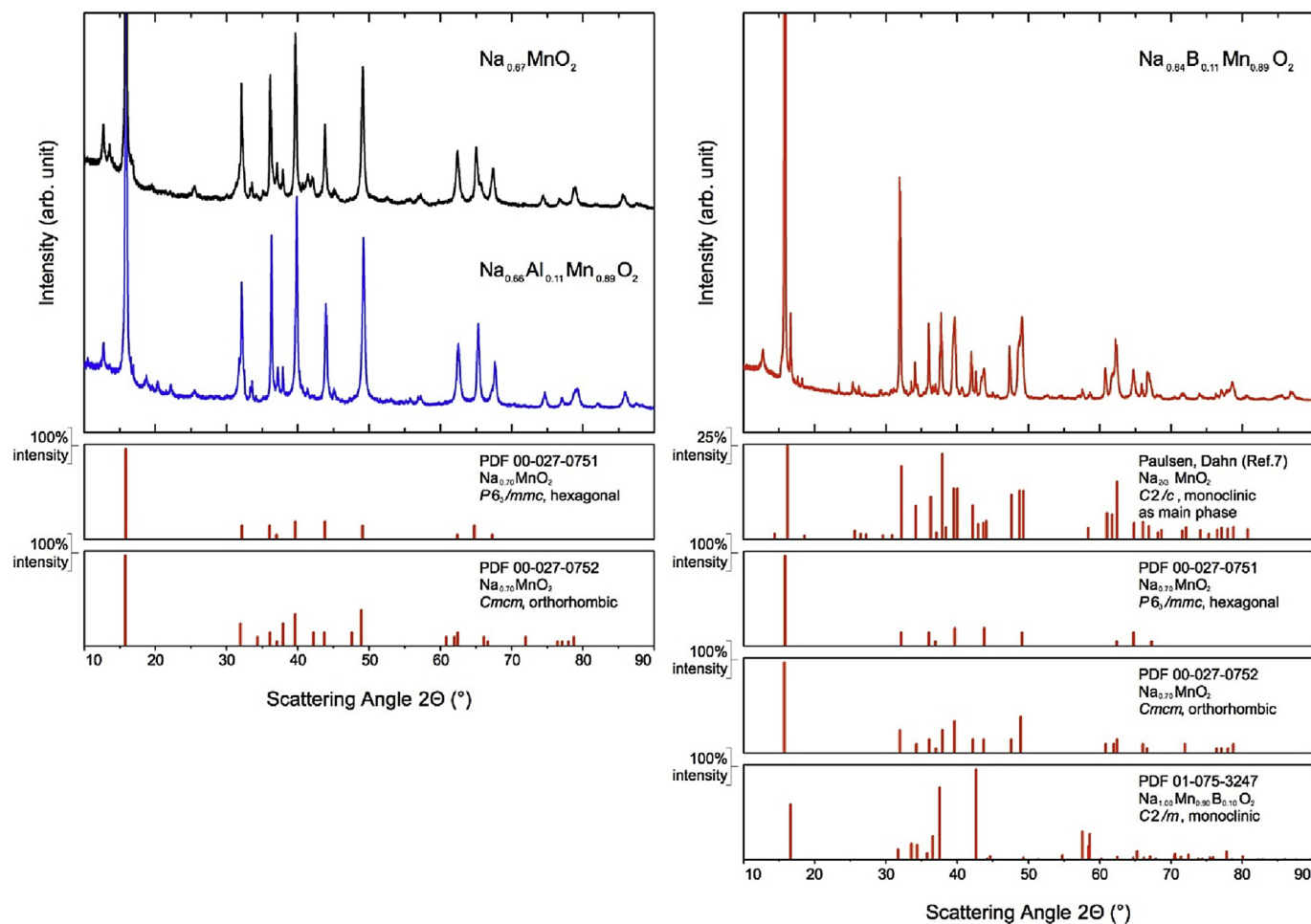


Fig. 2. X-ray diffractograms of $\text{Na}_{0.67}\text{MnO}_2$ (N_M), $\text{Na}_{0.66}\text{Al}_{0.11}\text{Mn}_{0.89}\text{O}_2$ (NAM) and $\text{Na}_{0.64}\text{B}_{0.11}\text{Mn}_{0.89}\text{O}_2$ (NBM). The diffraction patterns of relevant reference materials are also reported.

dQ/dV plot of P2- $\text{Na}_{0.6}\text{MnO}_2$ (space group $P6_3/mmc$) reported by Caballero et al. [23]. Both the cathodic and anodic sweeps are very reproducible with six main peaks present in each scan due to aforementioned reasons. The anodic current peak between 2.2 and 2.4 V is the most intense and only slightly decreases upon consecutive sweeps. Instead, the other anodic peaks at around 2.6, 2.8, 2.9, 3.5 and 3.7 V show a minor intensity fading. Correlating cathodic peaks are observed at around 2.1, 2.3, 2.5 and 3.5 V. Unlike for the anodic peaks, significant changes, i.e., potential shift and intensity decrease, are apparent upon the cathodic scan between 2.0 and 3.0 V, probably related to the active material degradation due to Mn dissolution [26]. Still, a good reversibility is observed for the material, especially in the 4.0–2.0 V range, with the main redox processes occurring at potentials below 3.0 V.

By expanding the cyclic voltammetry potential range to 4.6–1.5 V (see Fig. 3d), a significantly lowered reproducibility of the (de)sodiation processes is observed. In the first anodic sweep, an intense current peak occurs above 4.1 V, possibly related to electrolyte decomposition and partially irreversible phase transition [9]. As a matter of fact, a much lower intensity is measured in the following sweeps between 4.0 and 4.6 V. In addition, the anodic and cathodic peaks between 1.7 and 2.5 V show a much stronger shift and intensity decrease.

NAM, unlike the other materials in this study, shows a clear P2-like behavior in the CV (see Fig. 3b), similar to $\text{Na}_x\text{Mg}_{0.11}\text{Mn}_{0.89}\text{O}_2$ (space group $P6_3/mmc$) [3]. Between 4.0 and 2.0 V, the dominating

anodic and cathodic peaks occur at around 2.2 and 2.1 V, respectively. Other peaks appear at around 2.5, 3.5 and 3.7 V upon the anodic scan, but are much less intense. Overall, a minor shift and intensity decrease of the peaks is observed, indicating a very reversible cycling behavior. In the extended potential window of 4.6–1.5 V (see Fig. 3e), however, a much lower reversibility is detected. Significantly irreversible features are observed above 4.1 V as well as a large shift (0.3 V) of the main anodic peak.

In contrast to N_M and NAM, the CV of NBM (see Fig. 3c) shows significantly more features in accordance with the existence of multiple phases in this material (see Fig. 2). The most intense peaks occur at 2.4 and 2.3 V during the anodic and cathodic sweeps, respectively. However, a rather intense peak couple is seen to occur above 3.3 V upon the anodic scan which is similar to that reported in the dQ/dV plot of $\text{Na}_{2/3}\text{MnO}_2$ (phase mixture, space groups $Cmcm$ (70%) and $C2/n$ (30%)) [27], and in the dQ/dV plot of P2- $\text{Na}_{2/3}\text{MnO}_2$ (space group $Cmcm$) [25].

A lower reversibility of the (de)sodiation process is evidenced in the extended 4.6–1.5 V potential range (see Fig. 3f). A CV of NBM with 25 cyclic sweeps at 1.0 mV s^{-1} can be found in Fig. S8.

Overall, the CV results agree with the XRD analysis, as strong similarities to comparable materials with the identified phases are present. Conclusively, the CVs depict the electrochemical behavior of N_M and NAM to be similar to that of a typical P2-type layered oxide (i.e., $\text{Na}_{0.6}\text{MnO}_2$ and $\text{Na}_x\text{Mg}_{0.11}\text{Mn}_{0.89}\text{O}_2$; space group $P6_3/mmc$) whereas NBM is similar to that of $\text{Na}_{2/3}\text{MnO}_2$ (space group

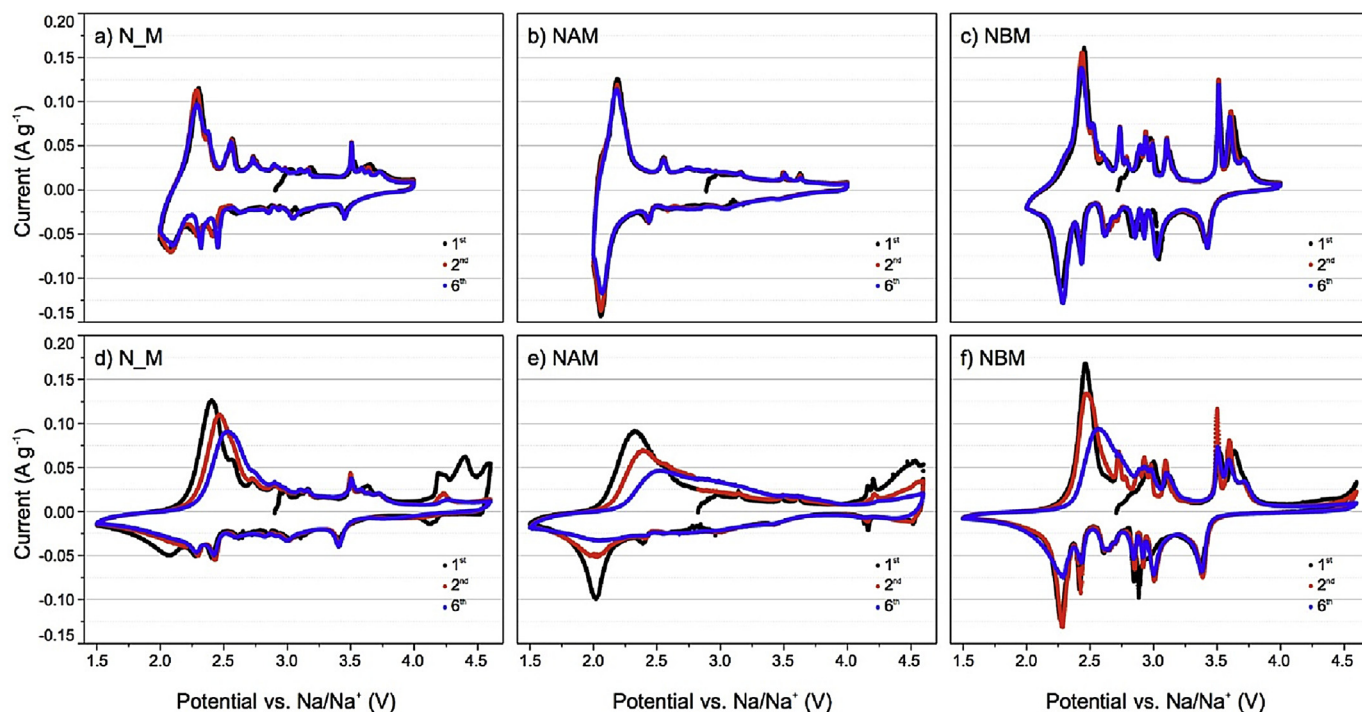


Fig. 3. Cyclic voltammetry profiles at 0.1 mV s^{-1} between 4.0 and 2.0 V (a–c) and 4.6–1.5 V vs. Na/Na^+ (d–f).

Cmcm). All materials reveal very good reversibility in the narrower potential range (4.0–2.0 V), but significant changes occur in the extended potential range (4.6–1.5 V).

Among the three materials, the highest peak intensities are observed for NBM, suggesting its superior electrochemical performance in terms of delivered capacity. In order to further confirm this hypothesis, galvanostatic cycling tests at constant current were performed for all materials in the potential ranges 4.0–2.0 V and 4.6–1.5 V at 10 mA g^{-1} or 40 mA g^{-1} . In this way, the cycling stability and reversible capacity of each material can be easily evaluated.

The first cycle potential profiles for the three materials are depicted in Fig. 4 with the equivalents of cycled Na^+ and the specific capacity (as calculated by Faraday's law based on the molar mass of $\text{Na}_{2/3}\text{X}_{0.11}\text{Mn}_{0.89}\text{O}_2$ with $\text{X} = \text{Mn, Al and B}$) in the two potential ranges. Additional potential profiles for the cycling at 10 mA g^{-1} can be found in Fig. S4 in the SI.

For N_M, an initial charge capacity of 53 mAh g^{-1} (0.20 eq. Na^+) and an initial discharge capacity of 150 mAh g^{-1} (0.57 eq. Na^+) are recorded in the range of 4.0–2.0 V (see Fig. 4a). The values increase to 128 mAh g^{-1} (0.49 eq. Na^+) upon charge and 186 mAh g^{-1} (0.71 eq. Na^+) upon discharge by setting the potential range between 4.6 and 1.5 V. It has to be noted that potential profiles and capacities are similar to those reported for $\text{Na}_{2/3}\text{MnO}_2$ (hexagonal, space group $\text{P6}_3/\text{mmc}$) [23,25].

The potential profiles for NAM (see Fig. 4b) are similar to these of $\text{Na}_x\text{Mg}_{0.11}\text{Mn}_{0.89}\text{O}_2$ [3]. The capacity in the first charge (37 mAh g^{-1} ; 0.14 eq. Na^+) and discharge (133 mAh g^{-1} ; 0.49 eq. Na^+) steps within the 4.0–2.0 V potential range are lower than those detected for N_M. However, increasing the potential range to 4.6–1.5 V, the material still offers a lower capacity during charge (120 mAh g^{-1} ; 0.44 eq. Na^+), but higher during discharge (195 mAh g^{-1} ; 0.72 eq. Na^+), as compared to N_M.

The potential profiles of NBM (see Fig. 4c) show several

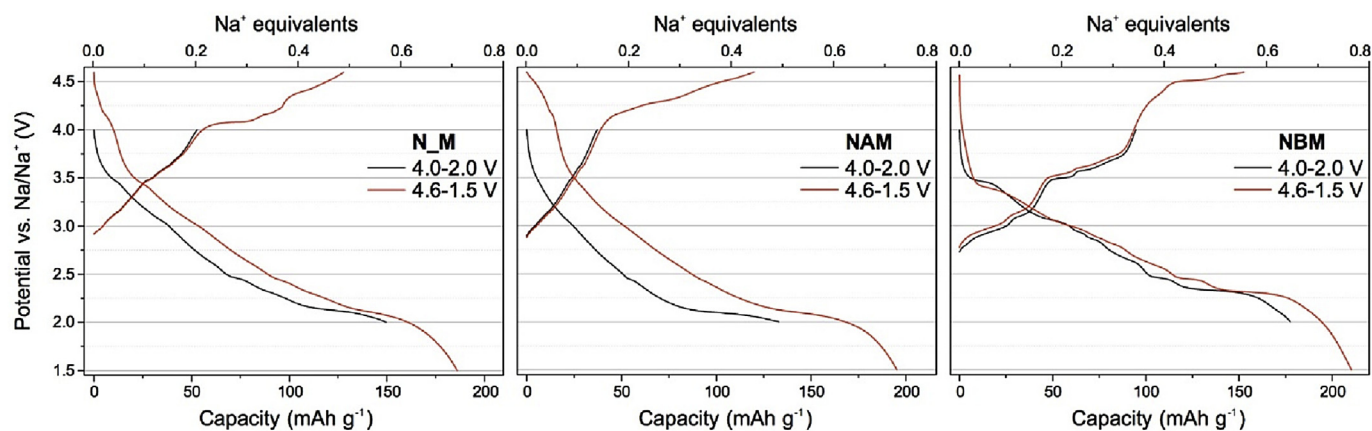


Fig. 4. Potential profiles of the first cycle at 10 mA g^{-1} between 4.0 and 2.0 V (black) and 4.6–1.5 V (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

individual steps, substantially more evident than in the other materials. This is remarkable, since most of the substituents are known to smooth the potential profiles of manganese-based compounds [27]. In addition, the recorded discharge capacities in both potential ranges are significantly higher than for N_M and NAM. At first charge, a capacity of 95 mAh g^{-1} (0.35 eq. Na^+) is recorded if cycled up to 4.0 V , followed by a delivered capacity of 177 mAh g^{-1} (0.64 eq. Na^+) during discharge to 2.0 V . Between 4.6 and 1.5 V , 0.56 eq. Na^+ (153 mAh g^{-1}) are extracted and 0.76 eq. Na^+ (210 mAh g^{-1}) are reinserted in the first cycle.

The long-term cycling behavior at 40 mA g^{-1} and selected potential profiles of the materials are shown in Fig. 5. All materials depict high coulombic efficiencies upon cycling within $4.0\text{--}2.0 \text{ V}$ (Fig. 5a), with NBM and NAM reaching more than 99.8% and N_M about 99.5% . The ratio between released and uptaken charge during the first cycle (which should not be confused with the coulombic efficiency) of NBM, NAM and N_M are 236% , 180% and 265% , respectively, due to the sodium deficiency of the layered materials, which highlights the need for developing an appropriate pre-sodiation in order to use such materials in practical sodium-ion cells. NBM reveals the best electrochemical performance and the highest recorded discharge capacity of 168 mAh g^{-1} . In contrast, the reversible capacities of N_M and NAM do not exceed 138 and 136 mAh g^{-1} , respectively. Moreover, NBM shows the highest capacity retention after 75 cycles (95%) as compared to N_M (86%) and NAM (79%).

In the wider potential range of $4.6\text{--}1.5 \text{ V}$ (Fig. 5b), increased

charge and discharge capacities are obtained. For N_M and NAM, a significant increase is evident, i.e., to 166 mAh g^{-1} and 168 mAh g^{-1} in the 2nd discharge process, respectively. In comparison, NBM shows a smaller relative capacity increase to about 178 mAh g^{-1} in the second cycle. However, the capacity fading increases and coulombic efficiencies decrease to about 99.2% for all three materials. As a matter of fact, the capacity retentions after 35 cycles is about 86% for NBM, 75% for N_M and 79% for NAM. In view of the lower reversibility, the improved capacity and electrochemical performance of NBM between 4.0 and 2.0 V appears to be very advantageous.

The potential profiles detected during the 2nd and 25th cycles at 40 mA g^{-1} between 4.0 and 2.0 V of N_M, NAM and NBM electrodes are shown in Fig. 5c. The potential profiles of NBM exhibit only very minor changes, which indicates a superior electrochemical stability. The other materials, N_M and NAM, show substantial fading between the 2nd and 25th cycles. The potential profiles in a wider potential range between 4.6 and 1.5 V at 40 mA g^{-1} (Fig. 5d) provide evidence for a poor cyclability of all materials. Additional potential profiles at 40 mA g^{-1} are depicted in Fig. S5.

Summarizing, in the $4.0\text{--}2.0 \text{ V}$ potential range, NBM shows much higher capacity and more stable cycling behavior (i.e., lower voltage and capacity fading) than the other materials. For that, the long-term cycling and rate performance in this potential range was investigated for NBM in greater detail (Fig. 6). The material shows a good performance although a capacity fade is present (Fig. 6a). In detail, the discharge capacities delivered within the $4.0\text{--}2.0 \text{ V}$

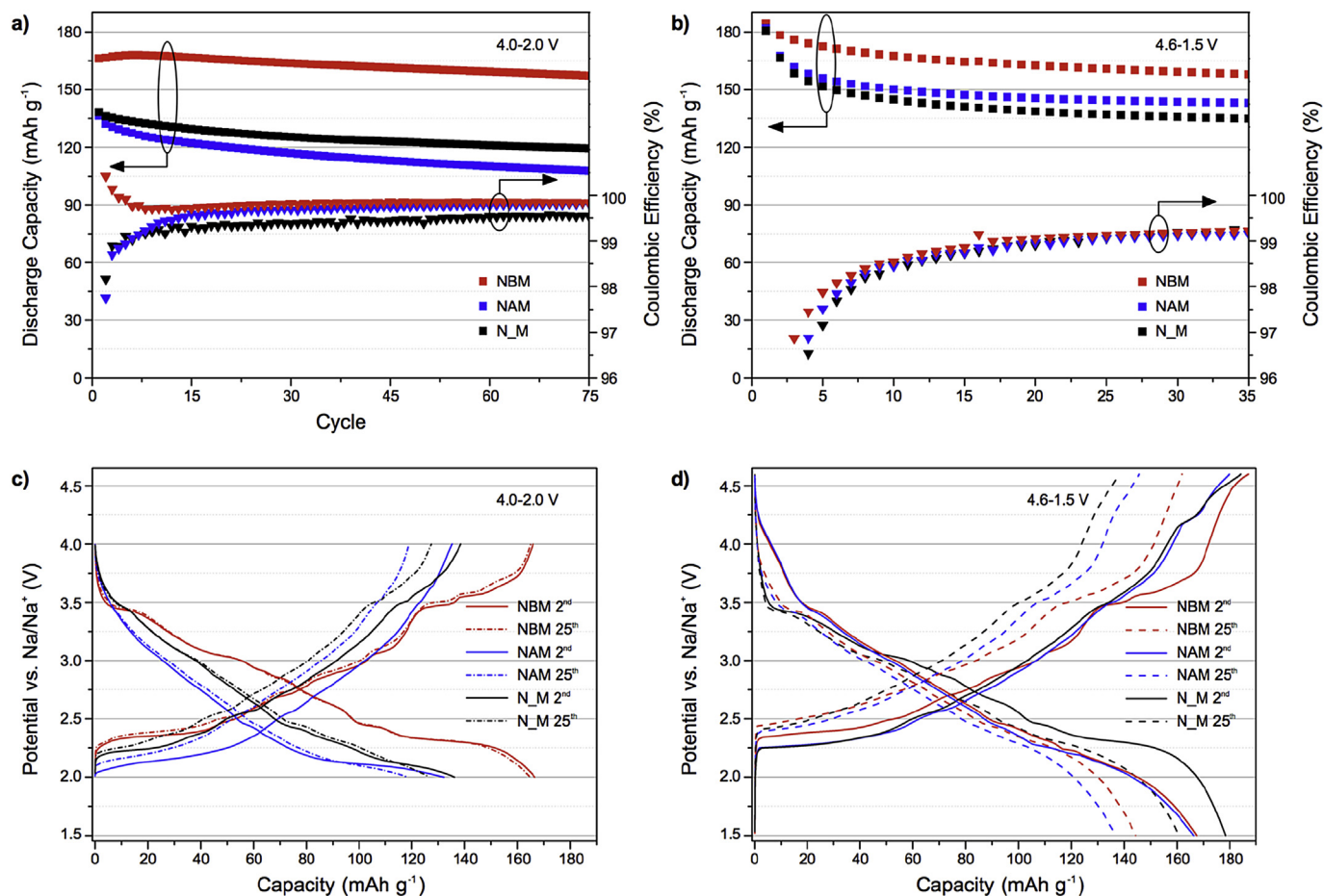


Fig. 5. Constant-current cycling with coulombic efficiency plots at 40 mA g^{-1} between a) $4.0\text{--}2.0 \text{ V}$ and b) $4.6\text{--}1.5 \text{ V}$ vs. Na/Na^+ . The corresponding potential profiles of cycle 2 and 25 are depicted for c) $4.0\text{--}2.0 \text{ V}$ and d) $4.6\text{--}1.5 \text{ V}$.

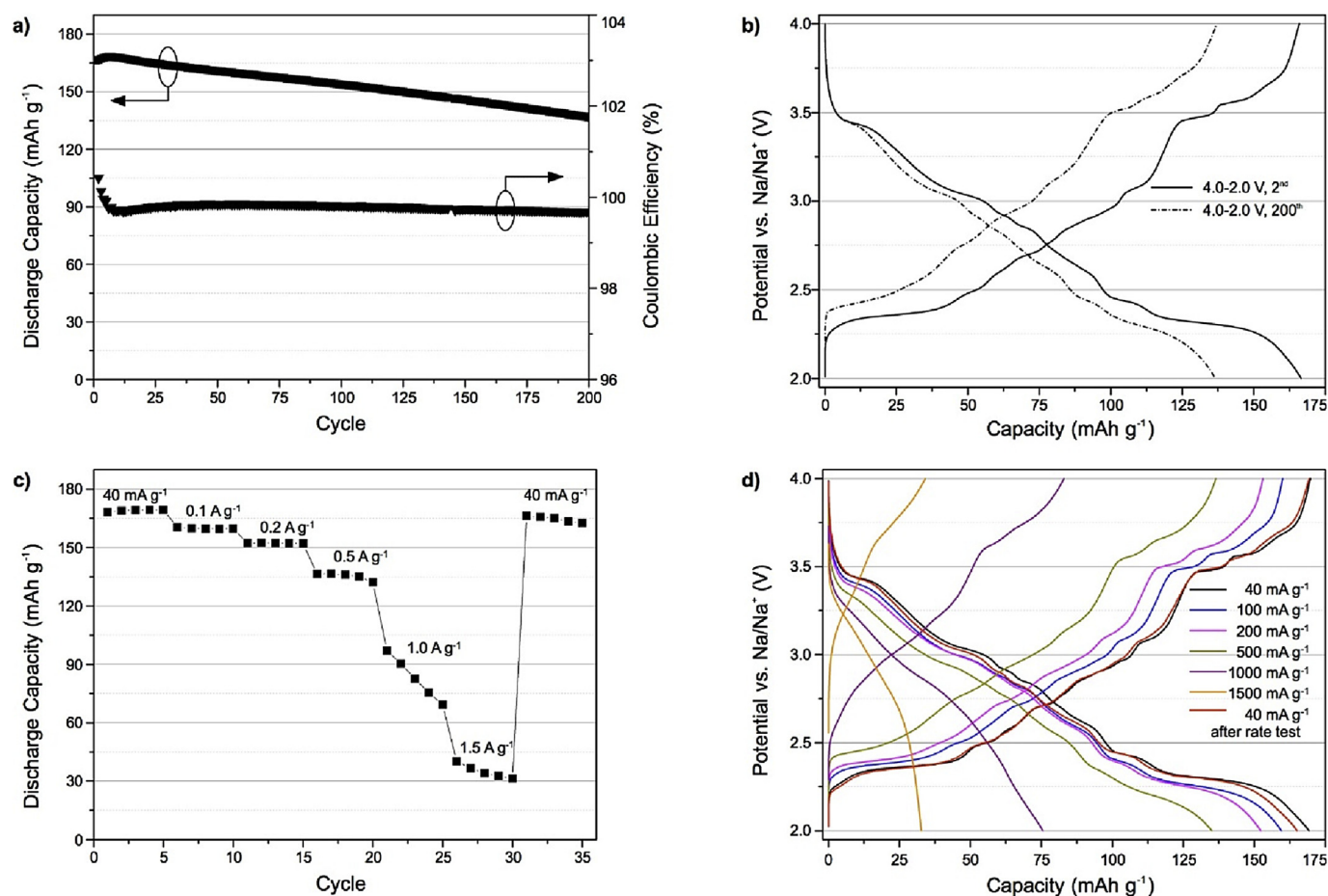


Fig. 6. a) Long-term constant-current cycling of NBM at 40 mA g⁻¹ between 4.0 and 2.0 V vs. Na/Na⁺ and b) selected potential profiles of the 2nd and 200th cycle for each potential range. c) Discharge capacities and d) selected potential profiles of NBM during a current-rate test at 40, 100, 200, 500, 1000 and 1500 mA g⁻¹ between 4.0 and 2.0 V.

potential range are 166, 161 and 137 mAh g⁻¹ at cycle 1, 50 and 200, respectively. This results in 82% capacity retention after 200 cycles. The coulombic efficiency remains high throughout cycling, reaching about 99.8% after 50 cycles. The long-term cycling of NBM in the range of 4.6–1.5 V is depicted in Fig. S6.

Concerning energy efficiency, the integrated potential profiles of NBM at 40 mA g⁻¹ in the range of 4.0–2.0 V show a very good performance. During the 2nd cycle, NBM features 481 Wh kg⁻¹ and 456 Wh kg⁻¹ during charge and discharge, respectively, corresponding to 94.8% energy efficiency. In the 200th cycle, the electrode features 415 Wh kg⁻¹ during charge and 376 Wh kg⁻¹ upon discharge, leading to an energy efficiency of 90.6%, i.e., only 4% lower compared to the 2nd cycle.

For a further analysis of the electrochemical performance of NBM, galvanostatic cycling testing at increasing currents was performed, as depicted in Fig. 6c. The discharge capacity during the initial five cycles at 40 mA g⁻¹ is around 165 mAh g⁻¹. Following, the charge and discharge currents are increased to 100, 200, 500, 1000 and 1500 mA g⁻¹ for five cycles at each current level. The material delivered rather high discharge capacities 160, 150 and 135 mAh g⁻¹ at 100, 200 and 500 mA g⁻¹, respectively. Indeed, the potential profiles (see Fig. 6d) reveal an increasing ohmic drop but do not depict a significant polarization, i.e., strongly sloping potential profile, indicating that the limitation is not due to the Na⁺ diffusion in the active material. However, upon charge and discharge at 500 mA g⁻¹, the material shows polarization in the consecutive 5 cycles, which is further increased at higher currents

(1000 and 1500 mA g⁻¹). Nonetheless, after such a stressing high-current test, the material does not show indications of material degradation, as proven by the almost overlapping potential profiles at 40 mA g⁻¹ recorded before and after the current-rate test.

4. Conclusion

Boron-doped layered sodium manganese oxide (Na_{2/3}B_{0.11}Mn_{0.89}O₂) was successfully synthesized via solid-state reaction. The material was thoroughly investigated in terms of structural and electrochemical properties, which were also compared with those of layered undoped and aluminum-doped manganese oxides. Adding boron results in the formation of different phases and, most important, a superior electrochemical performance compared to undoped Na_{2/3}MnO₂ and aluminum-doped Na_{2/3}Al_{0.11}Mn_{0.89}O₂. Na_{2/3}B_{0.11}Mn_{0.89}O₂ shows the highest discharge capacity (165 mAh g⁻¹) and capacity retention (82% after 200 cycles) upon long-term cycling at 40 mA g⁻¹. Additionally, the boron doping enhances the average potential upon discharge resulting in higher energy storage performance within the 4.0–2.0 V (vs. Na/Na⁺) potential range. The energy efficiency decreases by only 4% (starting from almost 95%) after 200 cycles. Also, the material depicts good rate performance with discharge capacities of 160, 152 and 135 mAh g⁻¹ at charge and discharge currents of 100, 200 and 500 mA g⁻¹, respectively. Finally, no structural degradation is evident after stressing the material at high current rates of up to 1500 mA g⁻¹.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2017.08.012>

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